

1. What is the role of the Provider component in React Context?

The **Provider** component makes data (state, functions, etc.) available to all components in the React tree **without having to pass props manually at every level**.

- It's part of the **Context API**.
- You wrap your component tree with the Provider and pass a value prop that can be accessed by any child component using useContext.

Example:

```
<MyContext.Provider value={user}>  
  
  <Profile />  
  
</MyContext.Provider>
```

Now, any component inside <Profile /> can access user directly.

In short:

The Provider “provides” context data to its child components.

2. How does the useReducer hook differ from useState?

Feature	useState	useReducer
Purpose	Used for simple state management	Used for complex state logic with multiple actions
State update	You directly set the new state	You dispatch an <i>action</i> that describes how to change the state
Structure	[state, setState]	[state, dispatch]
Best for	Simple counters, form fields	Complex apps, multiple related state transitions

Example:

```
// useState  
  
const [count, setCount] = useState(0);  
  
setCount(count + 1);  
  
// useReducer
```

```
const [state, dispatch] = useReducer(reducer, { count: 0 });

dispatch({ type: "increment" });
```

3. How do you dispatch actions to a reducer in React?

You use the **dispatch** function returned by useReducer to send an **action object** to the reducer.

Example:

```
function reducer(state, action) {

  switch (action.type) {

    case "increment":

      return { count: state.count + 1 };

    case "decrement":

      return { count: state.count - 1 };

    default:

      return state;

  }

}
```

```
const [state, dispatch] = useReducer(reducer, { count: 0 });
```

```
// Dispatching actions
```

```
<button onClick={() => dispatch({ type: "increment" })}>+</button>
```

```
<button onClick={() => dispatch({ type: "decrement" })}>-</button>
```

Here,

- dispatch() sends the action to the reducer.

- The reducer checks the `action.type` and updates the state accordingly.